

STATISTICAL UNCERTAINTIES OF END-POINTS IN GRAN PLOTS

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SUMMARY

Gran plots are commonly used to find equivalence points in acid–base and oxidation–reduction potentiometric titrations. It is shown that a standard error estimate for the equivalence point in a Gran plot can be calculated from the titration data by a weighted nonlinear optimization method. An example of the titration of a weak acid with a strong base is used to illustrate the calculational procedure.

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The class of data-evaluation methods for detecting equivalence points in potentiometric titrations which was proposed by Gran in 1952 [1] has since received wide attention [2–7] as an alternative to the more traditional methods which detect inflection points [8] in the titration curve. Another possibility is the computer fitting of model equations describing the titration to the titration data by nonlinear regression algorithms, as described in several papers [5,9–11]. A brief comparative discussion of these three classes of data evaluation serves as a useful introduction.

The inflection-point methods are the simplest in the sense that specific information about the nature of the analyte is not required. The titration curve is plotted in the form of pH (or potential) vs. titrant volume, this curve is visually examined for an inflection, and the volume coordinate of the inflection is obtained graphically or by a simple calculation. The same procedure applies to all analytes, strong and weak acids or bases, oxidizing or reducing agents and specific ion species. Obviously, if a titration curve has no inflection, these methods cannot be used. By contrast, the Gran plots and nonlinear regression methods both require some prior knowledge of the analyte, but neither requires the existence of an inflection. The original Gran plots [1] apply to data from a segment of the titration curve where a single reaction dominates in the analyte solution. A limited amount of prior information is needed: whether the analyte is a strong or weak acid or base or in redox titrations, the number of electrons in the analyte half-reaction. The Gran plots do not need values of acid–base equilibrium constants or redox formal potentials. The primary ad-

vantage of Gran plotting is the simplicity provided by the graphical data evaluation. In contrast, the nonlinear regression methods require a full knowledge of the titration equations and reasonable estimates of all unknown parameters. The details include mass and charge balances for all species involved in the titration, equilibrium-constant expressions or Nernst equations, varying activity coefficients and liquid-junction potentials. The principal advantage of nonlinear regression, in principle, is its capability of dealing with complicated titrations. This advantage is purchased at the cost of significant digital computing power. An analyst chooses from among these three classes of methods primarily on the basis of the amount of information that is available about the titration system and also on the calculational resources that can be devoted to the problem. There has been some effort to bridge the gap between the graphical simplicity of the Gran methods and the comprehensive nature of the nonlinear regression methods. These approaches modify the Gran equations in such a way that it is possible to use titration data derived from solutions in which the effect of the dominant reaction is perturbed by certain well-defined side-reactions. Much of the work in this area is related to titrations of seawater for alkalinity and carbonate content [12] but other more general side-effects have also been studied [6,13]. These modified Gran-plot methods require specific information about the side-effects to be compensated but do not require the complete set of model equations. The equivalence point for the dominant reaction can still be obtained graphically but some computer-aided procedures have been used [14].

This paper deals with one aspect of error evaluation for the equivalence point derived from the original (not modified) Gran plots. The uncertainty in this equivalence point, as with any experimental parameter, generally results from both random and systematic errors. The classification of error sources into these two categories is not always obvious and depends somewhat on the experimental procedure. Some errors are clearly random, e.g., electronic noise in a potentiometer circuit. Some errors are clearly systematic, e.g., an incorrectly adjusted pH-response slope. Other errors may have both random and systematic components. e.g., buret delivery volumes. Other errors are classified as random or systematic depending on if and how the titration is replicated. An example is the concentration of titrant delivered to the titration vessel. If the titration is done once, any deviation of the actual concentration from the "known" concentration is a systematic error. If the titration is replicated with the same solution of titrant, this error remains systematic and is thus not reflected in the standard deviation of the replicate equivalence-point volumes. But if a new solution of titrant is prepared and properly standardized for each replication, it is likely that the errors inherent in these titrant concentrations are random and will propagate randomly to the replicate equivalence points. In the face of this type of ambiguity in many aspects of a titration, the most rigorous (but unwieldy) procedure is to attempt to make all error sources random by replication with a completely different apparatus each time, including

potentiometers, electrodes, calibrating buffers, burets, analyte delivery devices and reagents for titrants. Clearly this is not normal practice. The rigorous estimation of the error in a potentiometric analysis does not usually justify the expense in time, effort and resources.

Studies of error in potentiometric titrations have largely dealt with the propagation of random error caused by the volume of titrant added and pH or potential measurements into the parameters calculated by nonlinear regression methods. Nonlinear regression algorithms routinely carry out this propagation along with the calculation of titration parameters. In addition, there have been a few studies of error evaluation by other methods of data evaluation. Meites et al. [15] studied the effect of random error sources on second-derivative inflection end-points. Hansson and Jagner [12] reported the effects of several sources of systematic error on titrations of seawater by both the original and modified Gran plots. Barron et al. [14] reported a comparative study of systematic errors in titrations of seawater using original and modified Gran plots and a nonlinear regression method. The current paper is believed to be the first to offer an approach to the problem of random error propagation within the original method for Gran plots. Systematic errors are propagated differently and so are not treated here. It is hoped that the method will be useful to those analysts who wish to use Gran plots and also require a random error estimate for the equivalence-point volume without doing replicate titrations.

ORIGINAL EQUATIONS FOR GRAN PLOTS

Different equations for Gran plots pertain to different types of titrations and to different branches of a given titration curve. For example, when an initial volume V of strong acid is titrated with a strong base of concentration C , the Gran equation for pH vs. titrant volume (v) data recorded before the first equivalence point, which occurs at $v=q$, is

$$(V+v)[H^+] = C(q-v) \quad (1)$$

However, if the analyte is a weak acid with conditional acidity constant K_a , the equation for data before the equivalence point becomes

$$v[H^+] = K_a(q-v) \quad (2)$$

For either strongly or weakly acidic analytes, the equation for data recorded after the last equivalence point q is

$$(V+v)[OH^-] = C(v-q) \quad (3)$$

If the analyte is a base and the titrant a strong acid of concentration C , then in the region before the first equivalence point, the Gran equations are

$$(V+v)[OH^-] = C(q-v) \quad (4)$$

if the base is strong, or

$$v[\text{OH}^-] = K_b(q - v) \quad (5)$$

if the base is weak with conditional constant K_b . After the last equivalence point in this case, the equation is

$$(V + v)[\text{H}^+] = C(v - q) \quad (6)$$

In all these examples, the hydrogen-ion concentration is calculated from the recorded pH by using the equation $\text{pH} = -\log[\text{H}^+]$ and the hydroxide ion concentration is calculated as $K_w/[\text{H}^+]$, where K_w is the conditional autoprotolysis constant. However, K_w need not be accurately known if the Gran plot is used solely for equivalence-point detection. These equations are all based on a number of simplifying assumptions [1] about the conditions of the titration reaction which reduce to the requirement that a single dominant reaction occurs.

For a simple redox potentiometric titration, in the region before the equivalence point q , the Gran equation is

$$v \exp(\pm n_A E/\alpha) = \exp(\pm n_A E'_A/\alpha) (q - v) \quad (7)$$

where $\alpha = RTF^{-1} = 0.02569$ at 25°C if the cell potential E is in volts and base e exponentiation is used; E'_A and n_A denote the formal potential and the number of electrons in the half-reaction of the analyte, respectively. The positive signs in the exponentials apply if the analyte is the oxidizing agent, the negative signs for a reducing analyte. After the equivalence point, the Gran equation is

$$\exp(\mp n_T E/\alpha) = (q^{-1}) \exp(\mp n_T E'_T/\alpha) (v - q) \quad (8)$$

where subscript T denotes the titrant couple. The negative signs in the exponentials apply if the titrant is reducing and the positive signs if the titrant is oxidizing.

In all these Gran equations, one plots the left-hand side vs. the titrant volume v and, if all the assumptions inherent in the derivations are satisfied, one expects to see a straight line which extrapolates to the horizontal v -axis at the desired equivalence point q . The left-hand sides are all functions of potentiometric measurements pH or E and, with the exception of Eqn. 8, also of the titrant volumes v . In general, the recorded potentiometric measurements and the recorded titrant volumes are both subject to random errors. The random error in the potentiometric measurements during the course of a given titration are due to instability in the pH or redox potential of the solution, to electronic fluctuations in the potentiometer circuitry and in the case of an analog potentiometer to reading error. Random errors, in volume addition derive largely from reading errors of buret graduations. Also if the buret tip is not in contact with the titration solution, the random volumes of hanging drops will also contribute error. In this development, it will be assumed that the potentiometric

random error is negligible and that the volumetric error is not. The same assumptions which lead to the above linear Gran equations also lead to conditions of stable potentials, i.e., well buffered acid-base solutions or well poised redox couples. Furthermore, with the proper use of a modern digital potentiometer and electrodes, there is virtually no circuit noise apparent in the digital display and there is no reading error. The neglect of potentiometric error simplifies the calculations required to derive the statistical uncertainties of the equivalence point in Gran plots. It is possible to treat cases of significant errors in both measurements provided that there is some prior information about the relative magnitudes of the two but this more complicated development is left to future research.

THEORY

The first aim is to put the eight Gran equations onto a common basis. The quantities V , K_w , n_A , n_T , and α are known constants and the quantities C , K_a , K_b , E'_A , E'_T , and q are unknown parameters. The recorded points to be used in the evaluation of the Gran plot are pH_i or E_i vs. v_i . It is clear that, in the presence of error, the measurements will not strictly follow any of the Gran equations. Rather these equations must be written in terms of error-free variables to be completely correct. Here the error-free titrant volume corresponding to v is denoted by the variable η and so each of the Gran Eqns. 1-8 is rewritten with v replaced by η . As discussed above, the potentiometric measurements are here taken to be free of random error so that these need no modification. A typical equation such as Eqn. 2 now, in terms of error-free variables, reads $[\text{H}^+] \eta = K_a(q - \eta)$. The left-hand side of each equation is of the form $h f(\eta)$. The functions $f(\eta)$ are $(V + \eta)$ in Eqns. 1, 3, 4 and 6, or η in Eqns. 2, 5 and 7, or unity in Eqn. 8. Thus the variables h , as the multipliers of $f(\eta)$, are $[\text{H}^+]$ or $[\text{OH}^-]$ or the redox exponential factor in Eqns. 7 and 8. The parameter b is defined as the slope of the straight line in the Gran plot, i.e., $\pm C$ or $-K_a$ or $-K_b$, etc. The parameter a is the value of $h f(\eta)$ at $\eta=0$, the intercept of the line with the vertical axis. With this common notation, the Gran equations become

$$h f(\eta) = a + b\eta \quad (9)$$

and the equivalence point in the Gran plot is

$$q = -a/b \quad (10)$$

If the random-error component of the recorded volumes is denoted by ϵ_i , these volumes are $v_i = \eta_i + \epsilon_i$, and the volume functions $f(\eta_i)$ are $(V + v_i - \epsilon_i)$ for Eqns. 1, 3, 4 and 6, or $(v_i - \epsilon_i)$ for Eqns. 2, 5 and 7, or unity for Eqn. 8. The variables h are error-free because they depend only on known constants and on poten-

tiometric measurements which are assumed to be error-free. Thus except for Gran plots derived from Eqn. 8, the common Eqn. 9 becomes

$$h_i f(v_i) - \epsilon_i h_i = a + b(v_i - \epsilon_i)$$

or with y_i defined as $h_i f(v_i)$

$$y_i = a + b v_i + (h_i - b) \epsilon_i \quad (11)$$

Equation 11 states that the y_i are linearly related to the corresponding v_i except for the error term $(h_i - b) \epsilon_i$. With y_i calculable entirely from recorded measurements and known constants, the adjustable parameters a and b can be fitted by using least-squares. However, all (y_i, v_i) pairs are not equally reliable because even if the volume errors ϵ_i are uniform, the presence of h_i in the error term introduces variability, i.e., heteroscedasticity. There is usually no evidence that the random error in the titrant volume varies appreciably during a given titration so that their common variance is taken to be σ^2 . The effective variances of the y_i become $(h_i - b)^2 \sigma^2 = w_i^{-1}$ and so each term in the sum is weighted by the weighting factor w_i . Thus, the weighted sum of squares to be minimized is

$$S = \sigma^{-2} \sum_{i=1}^n \left[\left(y_i - a - b v_i \right)^2 / \left(h_i - b \right)^2 \right] \quad (12)$$

This is not an ordinary weighted least-squares problem because of the presence of the slope b in the denominator. It is similar to, but not exactly the same as, the least-squares formulation that arises when both variables are subject to random error. This problem has been addressed by many authors, two of whom are York [16] and Williamson [17]. These authors make clear that the normal equations arising by setting the derivatives of S with respect to the parameters equal to zero are nonlinear in nature and must be solved numerically by an iterative calculation. Given the essential nonlinearity of the problem the option chosen here is not to derive normal equations from Eqn. 12 but rather to seek directly the two values of the parameters a and b which minimize the value of S as defined by Eqn. 12.

It should be noted that the ominous-looking differences in the denominators do not jeopardize the calculation. These denominators can never be zero. The h_i are always positive for all Gran equations except those derived from Eqn. 8. If the plot utilizes data before the first equivalence point, the slope b is always negative and so there is no problem. If the plot derives from Eqn. 3 or 6, the h_i and b are both positive but these quantities are both concentrations of the same species in the titrated solution and in the buret, respectively. Because of dilution, h_i must always be less than b . In the special case of Gran plots derived from Eqn. 8, the nonlinear formulation of Eqn. 12 is not necessary. In these cases, there are no h_i in the denominators because the y_i do not depend on v_i .

Hence, these particular cases can be solved by ordinary unweighted least-squares.

The calculational process which minimizes S with respect to the adjustable parameters a and b must also provide the variance and covariance estimates of those parameters at that minimum. These variances and covariances are necessary to estimate the statistical uncertainty of the equivalence point q as follows. Applying propagation-of-variance techniques [18] to Eqn. 10 leads to the relationship

$$\text{var } q = q^2 [(\text{var } a/a^2) + (\text{var } b/b^2) - 2 (\text{cov } (a,b)/ab)] \quad (13)$$

between the variance of the equivalence point and the variances and covariances of the parameters. The square root of $\text{var } q$ is the standard error estimate of the equivalence point in the Gran plot.

Many "optimization" methods are available for finding the minimum (or maximum) value of a function dependent on adjustable parameters. In this study, the simplex method, which has found extensive use in analytical chemistry [19] is adopted. This method has the advantage of not requiring evaluations of derivatives of the function with respect to the adjustable parameters. Furthermore, a derivation by Spendley [20] shows how to estimate the variances and covariances of the parameters from the final simplex configuration. Schwartz [21] has summarized the use of Spendley's method. The formulation given earlier [21] must be modified slightly for the weighting factors in the current problem. First in Appendix B [21], the matrix elements Ω_{pq} become $\sum_i w_i M_{pi} M_{qi}$ rather than the unweighted $\sum_i M_{pi} M_{qi}$. Secondly, the variance s^2 of the dependent variable y is identified with the variance σ^2 of the volume measurements. This quantity will not in general be known unless an independent statistical experiment is done to measure it. The usual practice [22] is to find the minimum weighted sum of squares in Eqn. 12 with σ^2 taken as unity. This minimum $S_{\min}$ is distributed σ^2 multiplied by the chi-square statistic with $(n-2)$ degrees of freedom. The expected value of $S_{\min}$ thus is $\sigma^2(n-2)$ and σ^2 is estimated by s^2 where $s^2 = S_{\min}/(n-2)$.

ILLUSTRATIVE EXAMPLE

One practical problem in the implementation of any Gran plot involves rejecting points in the region or regions of the titration curve in which the Gran plot assumptions are not valid. Typically a Gran plot in the appropriate branch of a titration appears as a straight segment with curvature at one or both ends of the branch. The curvature is caused by some violation of a Gran plot assumption. If the extrapolation to the horizontal v -axis is done graphically and by eye, it is a simple matter for the eye to ignore the curvature and to fit a straight line to the proper linear segment. However, if, as in the present case, a computer is needed to calculate the straight-line parameters, it is difficult to

program the computer also to reject the data points which deviate from an ill-defined straight segment. However, if the data in the curved regions are not excluded, not only will the equivalence point in the Gran plot be inaccurate but the statistical uncertainty calculated for that equivalence point will be overestimated if the variance of residuals is used to estimate σ^2 . The following illustrative example shows one way of proceeding, one that eventually relies on a judgement by eye for deciding the proper points to use in the analysis.

An aqueous analyte solution was prepared, containing about 1.6 mmol of potassium hydrogenphthalate (acidimetric standard) in 75.00 ml of 0.5 M potassium chloride. The KCl electrolyte serves to maintain virtually constant ionic strength so that the activity coefficients and liquid-junction potentials do not vary during the course of the titration. This analyte was titrated with standardized 0.0200 M sodium hydroxide in 0.5 M KCl from a 10-ml buret by using a Beckman model 71 pH meter which has a digital display reading to 0.001 pH. The pH meter was fitted with conventional glass and saturated calomel electrodes and was standardized with theoretical slope to pH 4.000 by using 1.00×10^{-4} M hydrochloric acid in 0.5 M KCl so that the meter would read pH in 0.5 M KCl as $-\log[H^+]$. The titration was done in a vessel thermostatted at $25.0 \pm 0.1^\circ\text{C}$. For this weakly acidic analyte, the appropriate Gran plot equation to apply in the region before the equivalence point is Eqn. 2. The recorded data for pH vs. titrant volume are plotted in the form $v[H^+]$ vs. v as shown in Fig. 1. This Gran plot exhibits pronounced curvature at titrant volumes less than about 3 ml and slight curvature at volumes above about 7 ml. The curved regions are explained by the fact that in the derivation of Eqn. 2 it is assumed that both $[H^+]$ and $[OH^-]$ are negligible relative to the concentrations of the weak acid and its anion. But in this case the analyte solution contains significant $[H^+]$ near the start of the titration and significant $[OH^-]$ as the titration approaches the equivalence point near $v=8$ ml. When either

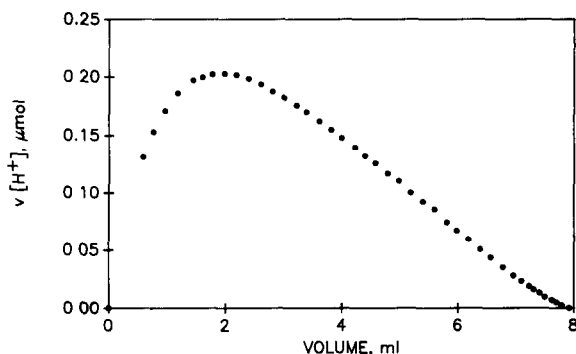


Fig. 1. Gran plot of the pH titration of a sample of potassium hydrogenphthalate solution with sodium hydroxide. Data from the curved segments may not properly be used in the estimation of the equivalence point or its statistical uncertainty.

$[H^+]$ or $[OH^-]$ is significant in any region of a weak acid titration, the Gran plot of the titration data does not follow the straight line predicted by Eqn. 2 and so the plot line is curved in that region. Schwartz [13] has shown how to modify the derivation to compensate for these effects and thus straighten out the resulting Gran plot, but those modified equations are not the subject of the current paper.

The illustrative example proceeds by provisionally rejecting the initial 14 data points corresponding to $v < 3.00$ ml and using the remaining 29 points to calculate the weighted least-squares Gran-plot parameters corresponding to the minimum S in Eqn. 12. This calculation yields an equivalence point q of 7.745 ml, a standard error estimate of q of 0.015 ml and a volumetric standard deviation s of 0.042 ml from Eqn. 14. The residuals of the data points from the fitted line are the numerators in Eqn. 12. These 29 residuals are plotted against the titrant volumes as shown in Fig. 2. If all the data points had been taken from a straight segment of the Gran plot, the residuals would have been scattered randomly above and below the horizontal zero line. However, the plot in Fig. 2 exhibits a sequence of residuals which is not random but rather has a pronounced sinusoidal pattern. The conclusion to be drawn is that some of the 29 points were taken from curved regions of the Gran plot. In the next step, several additional points, those below 4.40 ml and above 7.00 ml, were rejected in order to select only those points nearest the center of the straight segment. The remaining 14 points were used to calculate a new weighted least-squares Gran plot. The residuals about the new line are shown in Fig. 3. This plot is free of any apparent nonrandom pattern in the sequence of residuals and so one can conclude that the 14 central points were measured in analyte solutions

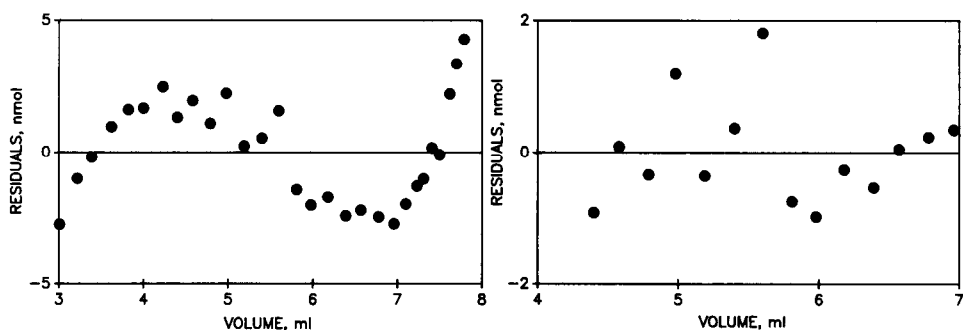


Fig. 2. Plot of the residuals of 29 data points from the titration with titrant volumes above 3.00 ml. The sinusoidal pattern is evidence that at least some of the points are derived from analyte solutions which do not conform to the assumptions for Gran plots.

Fig. 3. Plot of the residuals of 14 data points from the titration with titrant volumes between 4.40 and 7.00 ml. The random pattern of these residuals is evidence that all the data points are derived from analyte solutions which conform to the assumptions for Gran plots.

TABLE 1

Illustrative titration data and calculated results based on 14 points selected from the straight segment of Fig. 1

	v (ml)	pH	v (ml)	pH	v (ml)	pH
	4.40	4.523	5.40	4.768	6.39	5.102
	4.58	4.562	5.60	4.817	6.57	5.177
	4.79	4.614	5.81	4.894	6.78	5.284
	4.98	4.655	5.98	4.951	6.96	5.396
	5.19	4.715	6.18	5.017		
Parameter	a (mmol)		b (mmol ml ⁻¹)		q (ml)	
Value	3.138×10^{-4}		-4.111×10^{-5}		7.632	
Std. error	1.60×10^{-6}		2.68×10^{-7}		0.012	

Weighted standard deviation of residuals $s=0.0142$ ml

which conform to the assumptions for the Gran plot. The 14 points are listed in Table 1 together with the calculated least-squares parameters, the equivalence point, standard error estimates and the volumetric standard deviation s . The acceptable Gran line yields a standard deviation s of 0.014 ml which is significantly less than the provisionally calculated $s=0.042$ ml based on 29 data points. This latter value is invalid because it is inflated with systematic error caused by the lack of fit of the hypothesized straight line to the curved Gran plot. The volumetric standard deviation of $s=0.014$ ml is a more accurate measure of the random error in the titrant volumes delivered.

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